

KineSync-GS: Verified Visual State Synchronization for Articulated Gaussian Twins

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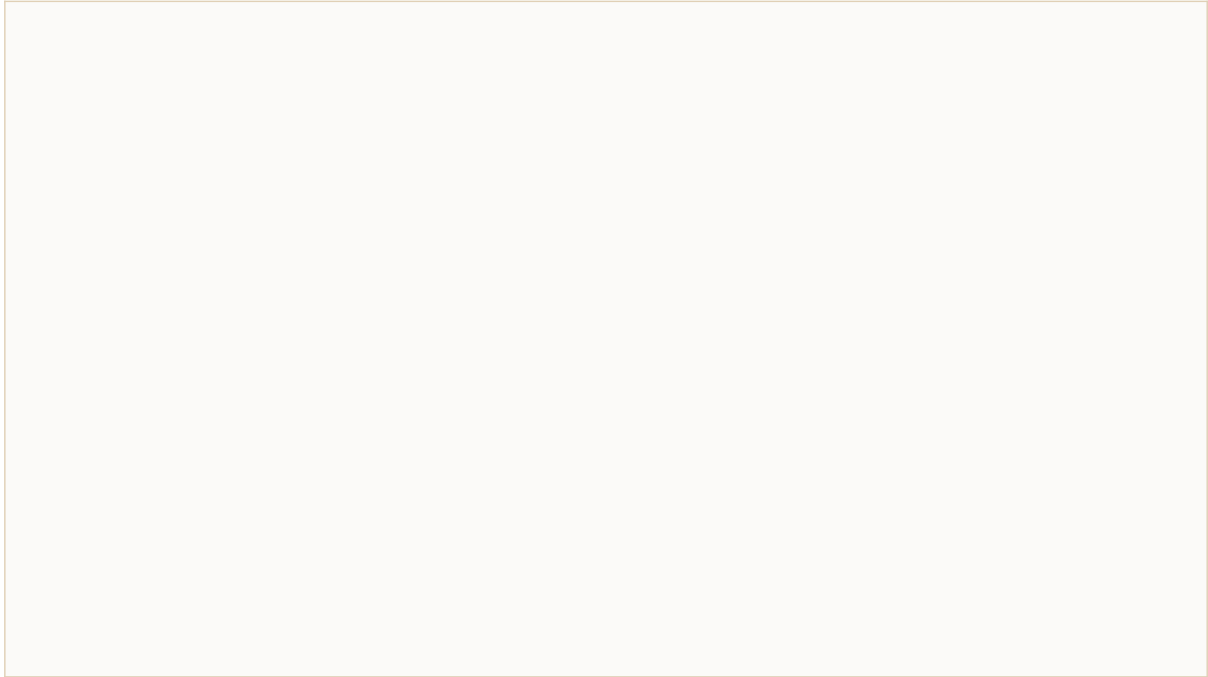


Fig. 1. KineSync-GS uses rendering or motion to propose a state correction, independent evidence to verify it, and a shared update/fallback interface to synchronize an articulated Gaussian twin.

Abstract—Gaussian robot twins can reproduce physical appearance with high fidelity, but their usefulness depends on synchronizing that appearance with the robot state that generated an observation. Direct visual fitting is insufficient: under inconsistent views, a renderer can lower image error by proposing an incorrect articulated state. We introduce KineSync-GS, a verified synchronization framework built around a single principle: an estimator proposes a correction, and independent evidence decides whether that correction may modify the twin. A kinematically bound Gaussian model produces bounded whole-state proposals, cross-view evidence verifies spatial updates, and motion evidence instantiates the same proposal–verification interface for camera–telemetry alignment. Unsupported proposals fall back to the measured state. On twelve held-out Franka poses, KineSync-GS reduces joint error from 0.590° to 0.322° while producing no harmful whole-state update across 48 conditions. In online PiPER experiments, the complete pipeline reaches 3.07 mrad with 0.36% harmful updates; supplying its synchronized state to an identical policy raises task-macro success from 58.33% to 71.67%. These results establish verified visual correction as a reliable state interface between real observations, articulated Gaussian twins, and robot policies.

I. INTRODUCTION

Photorealistic simulation increasingly connects real observations with embodied learning. Gaussian representations capture visual detail beyond manually textured geometry, while articulated models provide interaction-ready physical structure [1], [2], [3]. Together they form a visual twin when the rendered robot matches the observation in state and time. Telemetry alone cannot guarantee correspondence: joint-zero error, calibration residuals, stale views, and asynchronous streams can displace the rendered embodiment. Images directly expose these discrepancies, while link-bound Gaussians make them differentiable with respect to joint state.

A better rendering fit, however, need not imply a better state. Under disagreeing views, optimization can lower image loss by moving the twin away from the robot. Figure 2 exposes this failure mode: with a stale observation, visual loss falls by 50.02% while joint error reaches 5.073° . An independent decision must therefore control whether a fitted state may modify the twin.

We propose KineSync-GS, which separates correction

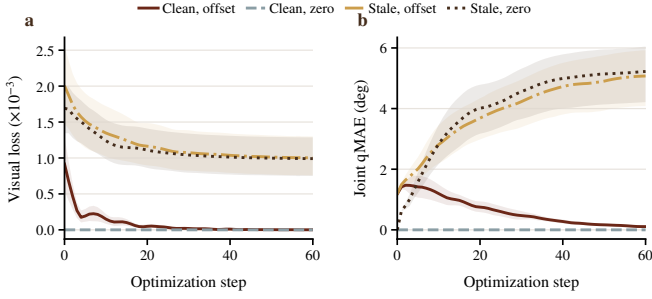


Fig. 2. Image fitting and state recovery can diverge. Curves aggregate all twelve held-out poses; bands show 95% pose-bootstrap intervals. Under stale views, optimization lowers visual loss while increasing state error.

proposals from state-changing decisions. A kinematically bound Gaussian renderer first estimates a bounded, coordinated joint correction. Independently recovered single-view estimates then provide evidence for accepting the complete proposal or retaining the measurement. The same scientific principle extends to time: image and joint motion propose a camera–telemetry offset, while calibrated motion evidence determines whether resampling is justified. Spatial and temporal synchronization thus follow one identifiable sequence: *proposal, evidence, verified update or fallback*. We test this principle through controlled Franka view conflicts, recorded-real PiPER retrieval across two Gaussian backends, trajectory-level timing disturbances, and the complete online PiPER pipeline with a fixed manipulation policy. Across these settings, the decisive improvement comes from regulating when visual evidence may change state rather than treating lower rendering loss as sufficient evidence.

Our contributions are: (1) A kinematically grounded Gaussian observation model that turns robot appearance into bounded whole-state proposals; (2) A unified proposal–evidence–update interface for spatial and temporal synchronization, including cross-view whole-state verification and measurement-preserving fallback; and (3) Paired controlled and recorded-real evaluations that connect state accuracy to update behavior across articulated assets, Gaussian backends, camera conditions, and timing disturbances, including online operation and same-policy manipulation on three real tasks.

II. RELATED WORK

3D Gaussian Splatting (3DGS) provides explicit primitives and differentiable rendering [1]. RoboGSim and GS-Playground combine it with closed-loop simulation [4], [3], while RoboSplat edits reconstructed Gaussians across visual worlds [2]. RE³SIM, GSWorld, and reconstructed soft-body simulation connect Gaussian appearance with physics-backed policy learning [5], [6], [7]; SplatCtrl updates scenes from RGB-D streams for reactive control [8]. These systems provide the substrate; KineSync-GS controls how visual evidence updates its articulated state.

Articulated recovery spans distance-model tracking [9], camera-to-robot estimation [10], [11], joint camera/robot refinement [12], and rigid pose estimation [13]. Differentiable Robot Rendering optimizes joints through Gaussian residuals [14]; GaussTwin couples physical prediction with visual

correction [15]; SplArt and active-view methods recover articulation or improve observations [16], [17], [18]. Our known skeleton instead produces bounded telemetry corrections whose complete admission is decided by independent cross-view evidence.

Temporal calibration jointly estimates sensor offsets and transforms [19], or recovers delay by aligning motion curves [20]. KineSync-GS correlates visual and joint motion, removes a calibrated native delay, and verifies the correction before resampling. Spatial and temporal paths share an estimate–verify–retain interface; OpenVLA-OFT [21] consumes the synchronized state without defining its estimator or update rule.

III. VERIFIED VISUAL STATE SYNCHRONIZATION

A. State and Observation Model

Let $q^m \in \mathbb{R}^J$ denote the measured arm-joint state and $\mathcal{C}$ the available camera views. For camera c , y_c is the image observation and ξ_c its calibrated parameters. The system estimates q^+ and produces a synchronized state q^s . A reference state q^* is used to calibrate decisions on development data and to score test predictions; it is not an input to test-time verification.

Each Gaussian k is attached to a robot link $\ell(k)$ with local mean $\mu_{k,0}$ and covariance $\Sigma_{k,0}$. Forward kinematics gives the link rotation $R_{\ell(k)}(q)$ and translation $t_{\ell(k)}(q)$, yielding

$$\mu_k(q) = R_{\ell(k)}(q)\mu_{k,0} + t_{\ell(k)}(q), \quad (1)$$

$$\Sigma_k(q) = R_{\ell(k)}(q)\Sigma_{k,0}R_{\ell(k)}(q)^\top. \quad (2)$$

Rendering these transformed Gaussians defines an observation $\mathcal{R}_c(q; \mathcal{G}, \xi_c)$. Physics and appearance therefore share the same articulated coordinates: the kinematic model constrains how the robot moves, and the Gaussian layer describes how that state appears.

Persistent calibration is separated from per-observation recovery. In the PiPER factorized model, training poses determine a shared joint-zero vector and one bounded SE(3) residual per camera, with the base world frame fixed. These quantities are held fixed when evaluating new state corrections. This separation prevents per-frame camera adjustments from absorbing the joint discrepancy that the state estimator is intended to recover.

B. Bounded Gaussian State Estimation

For selected joints I , let S_I insert a local correction into the full state. We optimize an unconstrained variable z through

$$\delta = b \tanh(z), \quad q^+ = q^m + S_I \delta, \quad (3)$$

where b specifies the correction bound. The visual objective averages observations over cameras:

$$\mathcal{L}(q) = \frac{1}{|\mathcal{C}|} \sum_{c \in \mathcal{C}} \ell_c(\mathcal{R}_c(q), y_c) + \lambda_p \ell_{\text{prior}}(q, q^m). \quad (4)$$

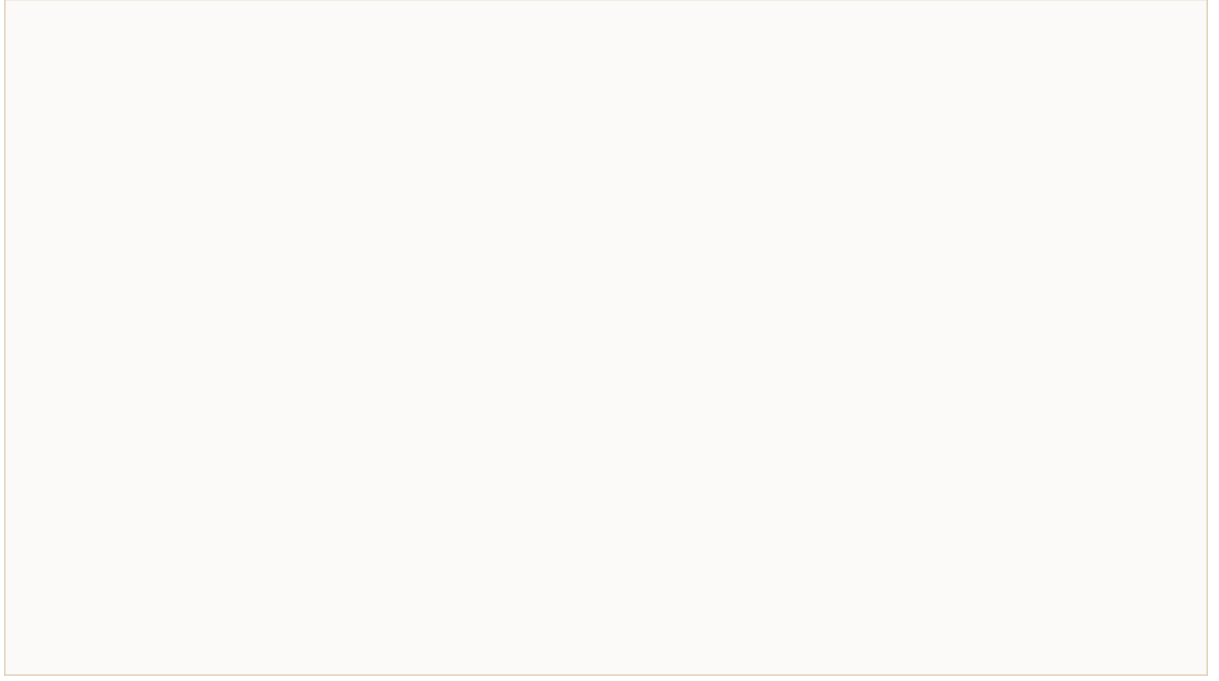


Fig. 3. Bounded Gaussian estimation followed by cross-view whole-state verification. Development calibration is fixed before evaluation.

The observation loss follows the available backend. PiPER supports soft-mask overlap, boundary alignment, and foreground RGB residuals with affine color fitting. The controlled Franka study uses mean RGB squared error over the two views. Both yield a bounded joint estimate through the same state interface; the verification stage is separate from the choice of photometric representation.

A fused estimate uses all views. We also recover δ_A and δ_B from the same starting measurement using each view separately. These single-view estimates are evidence about the fused estimate, rather than alternative action proposals. Their normalized disagreement is

$$d = \frac{\|\delta_A - \delta_B\|_2}{\max(\|\delta_A\|_2 + \|\delta_B\|_2, \epsilon)}. \quad (5)$$

When $\|\delta_A\|_2 + \|\delta_B\|_2 \leq \epsilon$, we set $d = 0$; we use $\epsilon = 10^{-8}$ throughout evidence computation. We measure visual improvement by

$$r = \frac{\mathcal{L}(q^m) - \mathcal{L}(q^+)}{\max(|\mathcal{L}(q^m)|, \epsilon)}. \quad (6)$$

For calibration, the evidence family additionally records gradient and correction-direction cosines,

$$g = \cos(P_I \nabla_q \ell_A(q^m), P_I \nabla_q \ell_B(q^m)), \quad (7)$$

$$a = \cos(\delta_A, \delta_B), \quad (8)$$

where P_I restricts both gradients to the same joint coordinates. Degenerate cosine denominators return zero. Evidence must be finite and use at least two views.

C. Development-Calibrated Whole-State Verification

We distinguish a visually improved estimate from a state update. For a development example i , define

$$e_i(q) = \frac{1}{J} \sum_{j=1}^J |q_j - q_{i,j}^*| \quad (9)$$

and label an estimate helpful when $e_i(q_i^+) < e_i(q_i^m) - \epsilon_e$, or harmful when $e_i(q_i^+) > e_i(q_i^m) + \epsilon_e$. We use $\epsilon_e = 10^{-8}$ rad; neutral examples contribute to neither label.

Let A_τ accept finite multi-view evidence when $r \geq \tau_r$, $g \geq \tau_g$, $a \geq \tau_a$, and $d \leq \tau_d$. We search development-derived thresholds by lexicographically minimizing

$$(N_{\text{harm}}(\tau), -N_{\text{help}}(\tau), -M(\tau), \tau). \quad (10)$$

Here N_{harm} and N_{help} count accepted harmful and helpful estimates. M is the minimum signed threshold margin over helpful and harmful development examples, rewarding separation after the two count criteria. A deterministic threshold order breaks remaining ties. This calibration uses labels only during development; test-time inputs are the measurement, estimated state, evidence, and fixed thresholds.

The final update is atomic at the state-vector level:

$$q^s = \begin{cases} q^+, & A_\tau(r, g, a, d) = 1, \\ q^m, & \text{otherwise.} \end{cases} \quad (11)$$

Thus an accepted correction preserves the coordinated estimate, and a rejected correction leaves the measurement unchanged. The estimator supplies the amplitude bound in Eq. (3); verification introduces no additional magnitude cutoff.

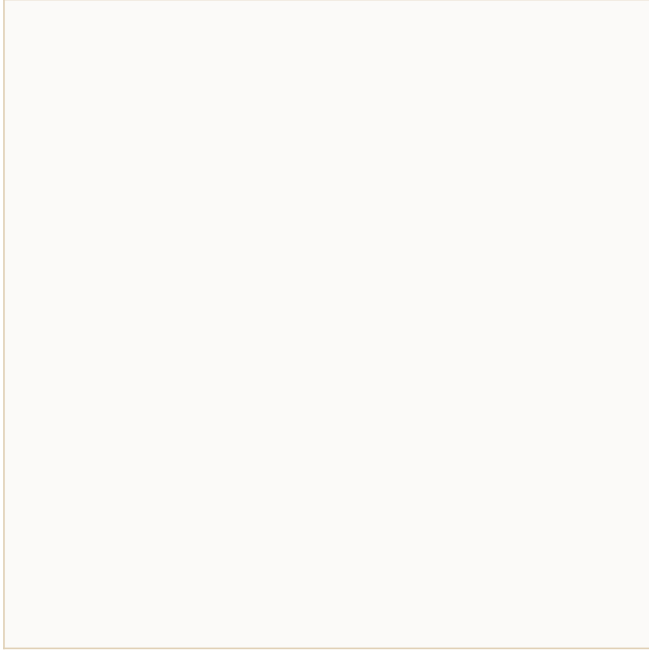


Fig. 4. Motion-based alignment of image and telemetry streams, followed by temporal verification and state resampling.

In the Franka development split, the selected thresholds are $\tau_r = 0$, $\tau_g = \tau_a = -1$, and $\tau_d = 0.42671$. The two cosine thresholds are nonrestrictive at this operating point. The effective checks are finite multi-view support, nonnegative visual gain, and bounded cross-view disagreement.

a) *Why preserve the complete estimate?*: Articulated visual residuals couple multiple joints. Masking individual components after joint estimation changes that coordinated solution. For intuition, consider a linearized least-squares residual $u + J_r \delta$ and a stationary minimizer δ^* . If D is a diagonal component mask, then

$$\begin{aligned} \frac{1}{2} \|u + J_r D \delta^*\|_2^2 - \frac{1}{2} \|u + J_r \delta^*\|_2^2 \\ = \frac{1}{2} \|J_r (D - I) \delta^*\|_2^2 \geq 0. \end{aligned} \quad (12)$$

The cross term vanishes by stationarity. This local identity explains why coordinate selection can discard a useful joint correction. It motivates verifying the complete estimate using external view consistency, rather than equating low residual with correct state.

D. Recorded-Real Visual Recovery

A second estimator instantiates the same estimate–verify interface using state-indexed Gaussian replay. Given a real PiPER head-camera image, we compare it with rendered states in a symmetric radius-four window around the delayed index in the selected frame sequence. Normalized RGB L_1 discrepancy selects the best matching state. A development-calibrated visual-gain threshold determines whether to use that state or retain the delayed input. A shared threshold of 0.00074274 is used for both Component-GS and Hybrid-GS.

This discrete estimator follows the same state contract as continuous whole-state recovery. Rendering proposes a state index, calibrated image evidence determines whether it may replace delayed telemetry, and rejection preserves

Algorithm 1 Cross-view whole-state synchronization

Require: Measured state q^m , views $\{y_c\}_{c \in \mathcal{C}}$, model $\mathcal{G}$, fixed thresholds τ

Ensure: Synchronized state q^s and verification record h

- 1: Validate calibrated views and initialize all estimators at q^m
- 2: $(q^+, \delta^+) \leftarrow \text{BoundedEstimate}(q^m, \{y_c\})$
- 3: **for all** $c \in \mathcal{C}$ **do**
- 4: $\delta_c \leftarrow \text{SingleViewEstimate}(q^m, y_c)$
- 5: $\ell_c^m, \ell_c^+ \leftarrow \text{RenderLoss}(q^m, q^+, y_c)$
- 6: **end for**
- 7: $(r, g, a, d) \leftarrow \text{CrossViewEvidence}(\{\ell_c^m, \ell_c^+, \delta_c\})$
- 8: $v \leftarrow \text{Finite}(q^+, r, g, a, d) \wedge |\mathcal{C}| \geq 2$
- 9: **if** $v \wedge A_\tau(r, g, a, d) = 1$ **then**
- 10: $q^s \leftarrow q^+$
- 11: $h \leftarrow (\text{ACCEPT}, r, g, a, d, \delta^+)$
- 12: **else**
- 13: $q^s \leftarrow q^m$
- 14: $h \leftarrow (\text{RETAIN}, r, g, a, d, \mathbf{0})$
- 15: **end if**
- 16: Record h with the fixed calibration identifier and input timestamps
- 17: **return** (q^s, h)

the input measurement. The estimator and evidence differ across backends; the proposal–verification–update interface does not.

E. Motion-Based Temporal Alignment

Images and telemetry also provide complementary motion cues for time alignment. We form a visual motion trace $v(t)$ from grayscale head/wrist frame differences and a joint-motion trace $s_\Delta(t)$ under a candidate offset. The temporal estimate is

$$\hat{\Delta} = \arg \max_{\Delta \in \Omega} \text{corr}(v(t), s_\Delta(t)), \quad \hat{c} = \hat{\Delta} - d_0, \quad (13)$$

where $\Omega = \{-200, -180, \dots, 200\}$ ms and d_0 is the native delay estimated from development zero-offset examples. A temporal verifier tests peak correlation, separation from alternative offsets, motion variability, and minimum correction magnitude. Accepted estimates resample the state stream; otherwise the output remains the original linear interpolation.

Spatial and temporal estimators therefore implement the same sequence: construct a correction proposal, evaluate it with independent evidence, and either update the complete state or preserve the input. Each path uses evidence matched to its observations while exposing the same synchronized-state interface to rendering, simulation, and downstream policies.

IV. EXPERIMENTAL PROTOCOLS

A. Evaluation Settings

We organize the evaluation around four synchronization questions (Table I): whether verified updates recover articulated state under controlled view conflicts, whether the interface transfers to recorded real observations and different Gaussian backends, and whether the same principle aligns image and telemetry streams and improves online manipulation. Figure 5 additionally confirms compatibility with the state interface of the public ABC policy and bimanual simulator [22].

TABLE I

EVALUATION PROTOCOLS. PERTURBATIONS REMAIN WITHIN THEIR LISTED BOOTSTRAP UNIT.

Protocol	Split	Held-out evaluation	Unit
Franka spatial	6 / 12 poses	4 conditions $\times$ 5 rules	Pose
PiPER visual	24 / 6 / 10 episodes	1,680 cases $\times$ 2 backends	Episode
Real temporal	4 / 2 trajectories	10 segments $\times$ 63 conditions	Trajectory
Online PiPER	9 / 27 sessions	3 tasks; 240 policy episodes	Session/pair

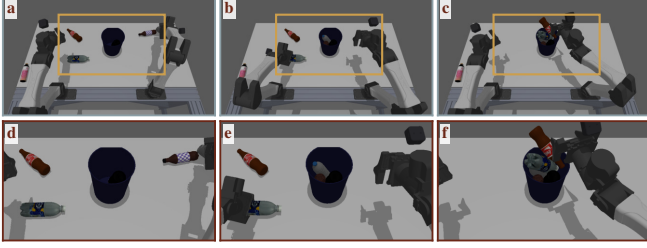


Fig. 5. ABC native-policy execution in the bottles-in-bin scene. Columns correspond to simulation steps 327, 981, and 1635 from seed 42002; the lower row enlarges the gold-boxed interaction region in each overhead view.

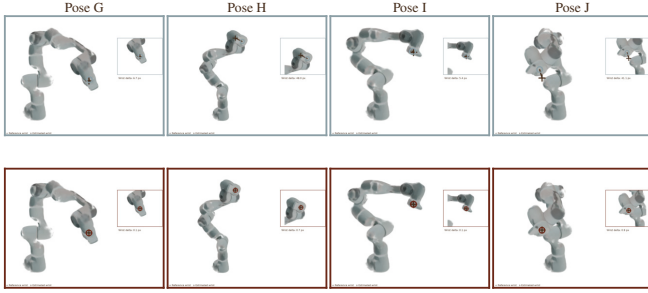


Fig. 6. Controlled Franka recovery on held-out poses G-J. Top: measured state. Bottom: KineSync-GS verified state.

a) Controlled spatial recovery.: We use the seven-joint Franka asset from GS-Playground [3] with two fixed views. Six poses calibrate verification; twelve nonoverlapping poses are held out. Each test pose is rendered under clean or stale-view observations with zero or nonzero measurement offset. All methods receive the same fused inverse estimate, bounded to 0.18 rad and optimized for 60 steps from 480×360 images. We compare measurement retention, unguarded inversion, global verification, component verification, and our whole-state verifier over 48 matched conditions. Figure 6 shows the measured and verified articulated states for four held-out poses.

b) Recorded-real visual recovery.: Forty PiPER red-cube episodes are separated into 24 training, six calibration, and ten test episodes. For each test image, Component-GS and Hybrid-GS render a symmetric state window around delayed telemetry. The methods are delayed state, unguarded retrieval, verified retrieval, and a paired-state oracle; all test decisions use thresholds fixed on the calibration split. The complementary real-robot sequences in Figs. 7 and 8 show the head/wrist observations and task progression represented by the recorded PiPER data.

c) Temporal synchronization.: The Take-Pens recordings contain synchronized head/wrist video and six-joint telemetry. Four files provide development data and two held-

out trajectories provide ten continuous test segments. We evaluate seven injected offsets under three jitter and three dropout levels, for 630 matched conditions. Raw nearest and linear interpolation, unguarded motion alignment, verified alignment, and an offset oracle share the same perturbed inputs.

d) Online real-robot evaluation.: The complete PiPER pipeline consumes synchronized head/external D455 images and six-joint telemetry in one online session. Eighteen held-out 60-second sessions cover three tasks and balanced disturbance segments; nine additional 20-minute sessions test continuous operation. Downstream evaluation holds the policy checkpoint fixed and changes only its state input over 240 paired episodes.

B. Metrics and Statistical Analysis

Joint qMAE averages absolute arm-joint error, reported in degrees for Franka and mrad for PiPER. A harmful spatial update increases qMAE over the input measurement; helpful utilization counts beneficial proposals that are applied. Visual recovery requires qMAE below 10 mrad. Temporal recovery additionally requires offset error below 20 ms. Update precision measures the fraction of applied corrections that improve state, while coverage measures how often a correction is applied when an error is present.

For online operation, end-to-end latency spans image acquisition through verified-state publication. Drift is measured during repeated pose holds, availability counts finite outputs within the state-age budget, and task success requires object placement and two seconds of stable release within 60 seconds. Franka intervals use 10,000 paired pose-cluster bootstrap replicates, with pose-level exact sign-flip tests and Holm correction for the primary comparisons. PiPER visual intervals use 2,000 episode-cluster bootstrap replicates. Temporal metrics are first aggregated within each held-out trajectory and then macro-averaged, keeping the trajectory rather than the perturbation as the independent unit.

V. RESULTS

A. Whole-State Verification Resolves View Conflict

Whole-state verification turns inverse rendering into a reliable state update. It obtains the lowest aggregate error in Table II, reducing qMAE from 0.590° for the measured state to 0.322° . Unlike the unguarded estimator, it uses visual corrections where the views agree and retains telemetry where their evidence conflicts (Fig. 9). The improvement follows from update selectivity rather than conservative inactivity. KineSync-GS applies 12 beneficial corrections while making no harmful update across all 48 held-out conditions. Figure 10 makes this operating behavior explicit: the method couples high helpful-proposal use with zero harmful updates, while Table II confirms zero drift from correct inputs.

Optimization traces explain why this separation is necessary. Under stale observations, visual loss continues to fall even as the proposed joint state reaches 5.073° error (Fig. 2). Verification evaluates the proposal using cross-view support after fitting, preserving the useful optimizer behavior

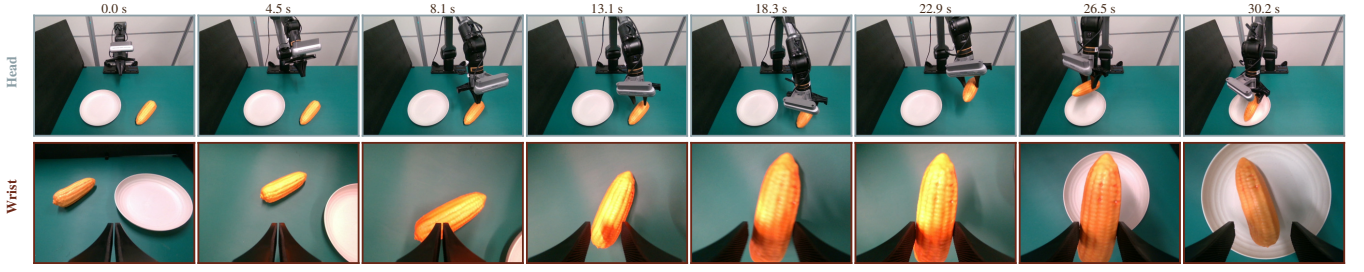
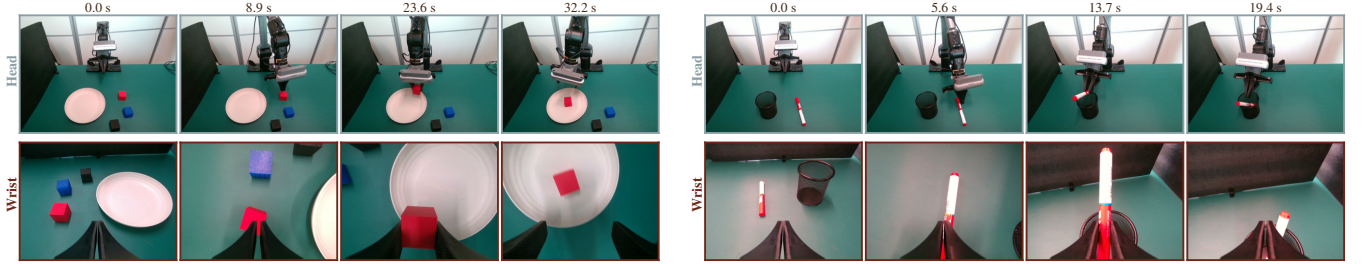


Fig. 7. Real-robot corn-to-plate experiment across matched head and wrist views. Eight temporal columns span the recorded PiPER execution from approach through placement.



(a) Red cube to plate (b) Red pen to bucket
Fig. 8. Real-robot PiPER experiments. (a) Red cube to plate and (b) red pen to bucket at four matched head/wrist times.

TABLE II

WHOLE-STATE RECOVERY ON 12 HELD-OUT FRANKA POSES. QMAE/DRIFT ARE IN DEGREES; BRACKETS ARE 95% POSE-CLUSTER BOOTSTRAP CIs.

Update rule	All qMAE [95% CI] ↓	Clean offset ↓	Stale offset ↓	Stale zero ↓	Harmful /48 ↓	Updates /48	Helpful /13 ↑	Zero drift ↓
Measured state	0.590 [0.566, 0.610]	1.180	1.180	0.000	0	0	0	0.000
Unguarded inverse	2.601 [2.100, 3.018]	0.106	5.073	5.224	23	36	13	2.612
Global verification	1.007 [0.683, 1.325]	0.106	3.921	0.000	8	21	13	0.000
Component verification	0.487 [0.418, 0.561]	0.465	1.482	0.000	6	20	12	0.000
Whole-state verification	0.322 [0.305, 0.337]	0.106	1.180	0.000	0	12	12	0.000

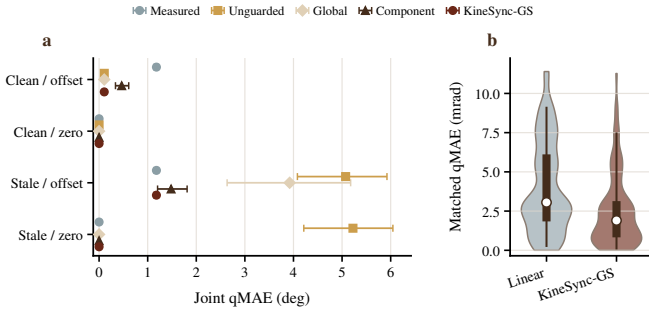


Fig. 9. Spatial and temporal recovery. (a) Franka joint qMAE with 95% pose-cluster bootstrap intervals across four view/offset conditions. (b) PiPER qMAE over 630 matched telemetry conditions; medians decrease from 3.05 to 1.90 mrad.

without equating its objective with state correctness. The same decision is visible in Gaussian geometry. For a held-out stale-view pose, unguarded fitting moves distal-link centers by 61.58 mm on average, whereas the verified state remains aligned at 7.70 mm (Fig. 11). The update rule therefore protects both the joint vector and the articulated Gaussian embodiment driven by it.

B. Verified Recovery Transfers to Recorded Real Data

The proposal–verification interface transfers across visual backends without changing its state contract. Starting from

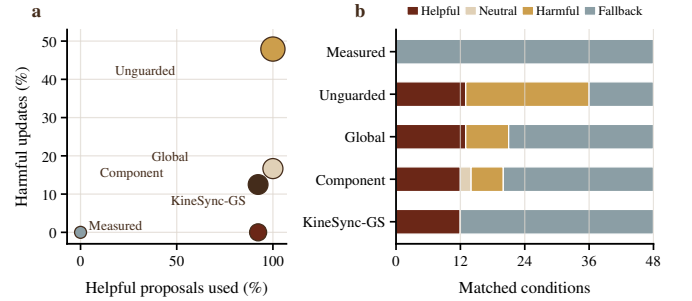


Fig. 10. Verified update behavior on the same 48 held-out conditions. (a) Helpful-proposal utilization versus harmful update rate; marker area encodes applied updates. (b) Update and fallback composition.

84.06 mrad delayed-state error, verified retrieval reaches 2.26 mrad with Component-GS and remains below 3 mrad with Hybrid-GS. Accepted states reduce gripper-region residuals rather than merely selecting a nearby index.

Verification also improves the quality of the updates that reach the twin. Across both backends, beneficial-update precision exceeds 94% while exact retention of zero-offset inputs rises over unguarded retrieval. This pairing of recovery and input preservation is the recorded-real counterpart of the controlled Franka result. The same design principle improves temporal synchronization under every tested jitter/dropout combination. Relative to linear interpolation, KineSync-GS

TABLE III

ONLINE PIPER SYNCHRONIZATION, SAME-POLICY MANIPULATION, CONTINUOUS OPERATION, AND VERIFIED-UPDATE MECHANISMS. QMAE AND DRIFT ARE IN MRAD AND MRAD/MIN; RECOVERY, HARMFUL UPDATES, FALLBACK, STABILITY, AND AVAILABILITY ARE PERCENTAGES.

Online synchronization over six held-out sessions per task						
Task	Online output	qMAE ↓	Recovery ↑	Harmful ↓	Fallback	
Take-Pens	Raw state	12.84	38.26	0.00	—	
	Temporal only	6.73	62.18	0.83	41.06	
	Combined, unguarded	3.84	81.34	7.48	2.76	
	KineSync-GS	3.46	79.62	0.42	22.36	
Red-Cube-to-Plate	Raw state	11.26	43.71	0.00	—	
	Temporal only	5.88	67.84	0.76	30.27	
	Combined, unguarded	3.28	85.21	6.62	2.14	
	KineSync-GS	2.94	84.36	0.36	19.84	
Corn-to-Plate	Raw state	10.92	46.18	0.00	—	
	Temporal only	5.64	69.12	0.69	26.68	
	Combined, unguarded	3.06	86.44	5.94	2.11	
	KineSync-GS	2.81	85.18	0.31	18.62	
Matched module ablations, macro-averaged over the three tasks						
Variant	Temporal	Gaussian	Verification	qMAE ↓	Recovery ↑	Harmful ↓
Raw state	No	No	None	11.67	42.72	0.00
Temporal only	Yes	No	Temporal	6.08	66.38	0.76
Spatial only	No	Yes	Whole-state	6.21	64.17	0.91
Combined, unguarded	Yes	Yes	None	3.39	84.33	6.68
Component update	Yes	Yes	Component	3.82	76.48	1.37
KineSync-GS	Yes	Yes	Whole-state	3.07	83.05	0.36
Same-policy task success: 40 paired episodes per task and state input						
State input	Take-Pens SR (count)	Red-Cube SR (count)	Corn SR (count)	Task-macro SR (%)	Success time (s) ↓	
Raw state	45.00 (18/40)	62.50 (25/40)	67.50 (27/40)	58.33	16.42	
KineSync state	62.50 (25/40)	75.00 (30/40)	77.50 (31/40)	71.67	14.86	
Three uninterrupted 20-minute sessions per task						
Task	E2E p50/p95 (ms)	Fresh corrections/s	Drift (mrad/min) ↓	Stable/available (%)	Fallback (%)	Completed
Take-Pens	486.73/791.46	1.36	0.18	98.64/98.72	24.38	3/3
Red-Cube-to-Plate	421.58/704.32	1.48	0.12	99.16/99.08	21.47	3/3
Corn-to-Plate	453.26/748.91	1.43	0.15	98.92/98.91	22.81	3/3
Cross-view calibration factorization						
Calibration	Cases	Recovered (count)	Recovery	Gain (pp)	Dual-view	Win/loss
Unfactorized	45	23/45	51.11	—	42.86	—
Factorized	45	29/45	64.44	+13.33	80.95	9/1
Component-level re-verification on matched candidates						
Update rule	Cases	Successful candidates	Retained (count)	Retention	Precision	Coverage
Magnitude gate	24	18	11/18	61.11	100.00	45.83
Component re-verification	24	18	16/18	88.89	100.00	66.67

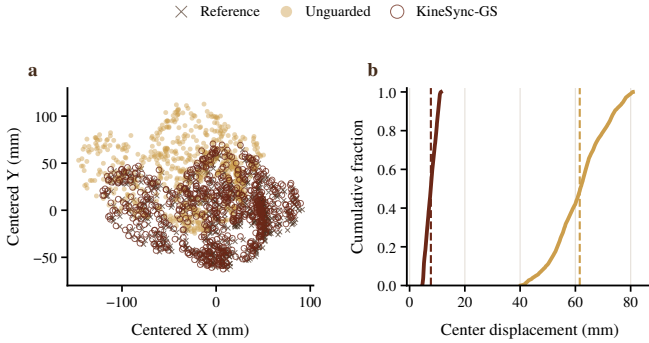


Fig. 11. Distal-link geometry under a stale view. (a) XY projections of corresponding Gaussian centers. (b) Empirical displacement distributions; dashed lines mark means.

lowers trajectory-macro error by 38.74% and reaches 97.97% beneficial-update precision. Figure 9b shows the resulting distribution shift over all 630 matched timing conditions, complementing the controlled spatial comparison in its left panel.

C. Independent Evidence Produces the Operating Point

Factorized calibration makes cross-view evidence more discriminative by separating shared joint-zero error from camera-specific residuals. On paired camera perturbations,

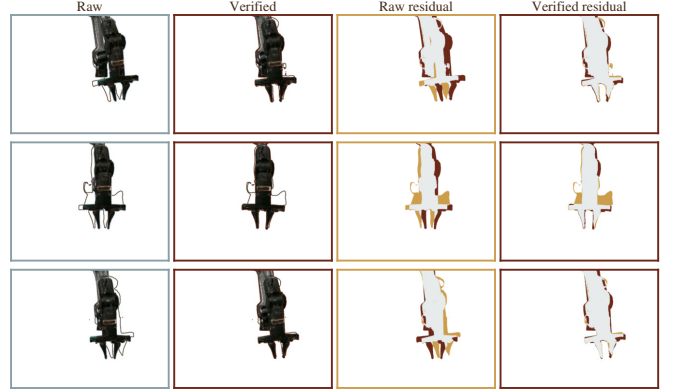


Fig. 12. Recorded-real PiPER alignment on three matched cases. Columns show measured and verified gripper regions and residuals; gold/brown indicate missed/excess foreground.

recovery rises from 42.86% to 80.95% (Table III), supporting the use of independently calibrated views as evidence rather than interchangeable render losses. Component re-verification complements this calibration by rechecking the proposed state before application. It retains 16 of 18 successful candidates, compared with 11 for magnitude gating, while preserving perfect precision in the matched analysis. At the image level, the same sequence is observed state, ren-

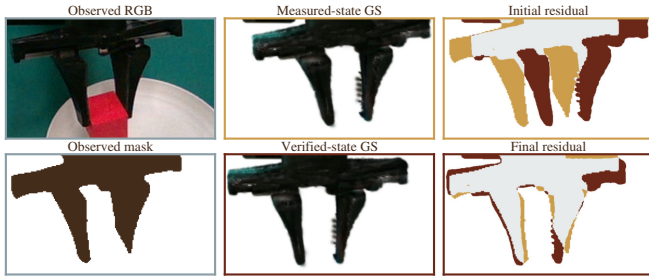


Fig. 13. Saved-state spatial diagnostic workflow. A recorded image and its target mask provide independent evidence for evaluating the measured-state render, proposed correction, and verified residual before the synchronized state is committed.

dered proposal, independent residual evidence, and verified commit.

The synchronized state is exposed through a robot-independent action interface. Six PiPER joints and a normalized gripper value map to both physical and Gaussian backends, and an OpenVLA-OFT endpoint [21] consumes the resulting state to produce finite action chunks. This interface keeps the synchronization decision separate from the downstream policy while providing the policy with one verified state representation.

D. Online Synchronization Improves Real Manipulation

The complete online chain preserves the same proposal-evidence-update contract while operating on newly arriving camera and telemetry streams. Across all three tasks, KineSync-GS reaches 2.81–3.46 mrad qMAE and limits harmful updates to 0.31–0.42% (Table III). Its recovery remains within 1.72 percentage points of unguarded fusion while reducing harmful updates by more than an order of magnitude. Matched module removal confirms that temporal and Gaussian evidence are complementary. Either path alone remains above 6 mrad, whereas their verified combination reaches 3.07 mrad and 83.05% recovery with only 0.36% harmful updates (Table III). The synchronized state also improves manipulation under an unchanged policy. Over 240 paired real-robot episodes, task-macro success rises from 58.33% to 71.67%, while successful completion time falls from 16.42 to 14.86 s. All nine 20-minute sessions complete without restart, with at least 98.72% output availability and at most 0.18 mrad/min drift (Table III).

VI. CONCLUSION

We introduced KineSync-GS, a verified synchronization framework for articulated Gaussian twins. Its central contribution is to separate state estimation from state-changing authority: rendering or motion proposes a correction, independent evidence evaluates that proposal, and the twin receives either a verified whole-state update or the original measurement. This common interface unifies spatial recovery from multi-view Gaussian observations with temporal alignment of camera and telemetry streams. Controlled Franka experiments show that this separation preserves useful inverse-rendering corrections while rejecting state changes caused by inconsistent views. Recorded-real PiPER results carry

the same behavior across multiple manipulation tasks, two Gaussian backends, and perturbed timing conditions. Online experiments further show that the verified state improves success for an unchanged policy while sustaining continuous operation across three tasks. The resulting combination of lower state error, precise updates, measurement-preserving fallback, and stronger manipulation establishes visual evidence as more than a rendering signal: it becomes a principled synchronization layer between robot observations, articulated Gaussian representations, and downstream policies.

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